Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/20/2026
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Cognitive and Physiologic Responses in Commercial Space Crew on Short-Duration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	basner@pennmedicine.upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-573-5866
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Psychiatry, Division of Sleep and Chronobiology		
PI Address 2:	423 Service Dr, 1013 Blockley Hall		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	12/01/2021	**End Date:**	03/31/2028
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	End date changed to 03/31/2028 per TRISH (Ed., 3/4/25). End date changed to 03/31/2025 per TRISH (Ed., 12/23/24). End date changed to 11/30/2024 per TRISH (Ed., 10/26/23). Previous end date was 11/30/2023.		
Key Personnel Changes/Previous PI:	Christopher Jones, Ph.D. added as CoI 2024		
COI Name (Institution):	Jones, Christopher Ph.D. (University of Pennsylvania Health System)		
Grant/Contract No.:	NNX16AO69A-EXP0009		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	October 2023 Update: The project was amended to collect data on SpaceX Polaris Dawn mission before, during and after a 5-day mission that includes an extravehicular activity (EVA), and Axiom's Ax-2 mission that will launch four spaceflight participants onboard a SpaceX Crew Dragon vehicle from Cape Canaveral, FL. This mission will dock to the International Space Station (ISS) United States Operating Segment (USOS). Once on orbit, the crewmembers will stay for approximately two weeks. The proposed research will be performed on two commercial crew flights: (1) The first one is performed by Space Adventures and includes one Japanese crew member who will launch on a Soyuz rocket from Baikonur in December 2021. He will spend a few days on the International Space Station (ISS) before his return to Russia. Data collection will be restricted to pre- and post-flight. (2) The second one is performed by Axiom Space and includes a crew of 4 (including one former female astronaut) who will launch on a Falcon rocket from Kennedy Space Center in December 2022. They will spend a few days on the International Space Station before they return to Florida. Data collection will be restricted to pre- and post-flight, but may include tests while docked to the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project demonstrates that commercial crews and Antarctic winter-over crews can collect valuable and valid cognitive and physiological data with relatively little effort. These data will help improve health and safety for future crews.
Task Progress:	This research supports the collection and analysis of survey, cognitive test, and physiological data in commercial space crews. So far, we have successfully supported 8 missions (Inspiration4, Space Adventures, Axiom-1, Axiom-2, Axiom-3, Polaris Dawn, Axiom-4, and Fram-2). We have also expanded this research to the space-analog environment of Antarctic Research stations in collaboration with the Australian Antarctic Program (AAP). Here, we have implemented monthly physiological measurements with BioIntelliSense's BioButton and cognitive testing with the Cognition test battery in four AAP research stations for two consecutive winter-overs. The primary objective of this research is to collect data on crew behavior and performance before, during (if possible) and after the mission, to extract the data, clean and annotate the data, and submit the data to the Translational Research Institute for Space Health (TRISH) Enhancing eXploration Platforms and ANalog Definition (EXPAND) database. We collect survey and cognitive performance data with NASA's Cognition test battery (developed for NASA by the Principal Investigator/PI). We originally used an Apple Watch to collect physiologic crew data, but have recently switched to Garmin watches, primarily due to the better battery life and better support (including custom-made firmware) by the watch manufacturer. Starting with Polaris Dawn, we have added BioIntellisense's BioButton as a physiological measurement tool. The BioButton is a single electrode system that captures various physiological signals (e.g., heart rate, respiratory rate, body temperature) but is not primarily designed for research, which requires extensive testing before the system can be successfully deployed in research and (especially) spaceflight settings. Lessons learned and data analyses are annually presented at the NASA Human Research Program (HRP) Investigators' Workshop (IWS) and potentially other intramural and extramural meetings. For example, we presented at Garmin's Health Summit in Singapore in November 2023 and in New York City in September 2025. In 2024, the flagship Inspiration4 manuscript was published in the journal Nature, with PI Mathias Basner as co-corresponding senior author. We have debriefed the Polaris Dawn and Fram2 crews in early 2025 and 2026, respectively.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Awards	Basner M. "Exceptional Scientific Achievement Medal by the National Aeronautics and Space Administration (NASA), July 2025." Jul-2025